Group 2

AI x HR Hackathon

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Context

LuminaTech Industries

- Manufacturing electronic components for "Industry 4.0".
- Build the basic hardware for smart factories
- Products : electronic sensors, circuit boards, and control panels used in automated assembly lines and robotic arms

Industrial Electronics/ Manufacturing

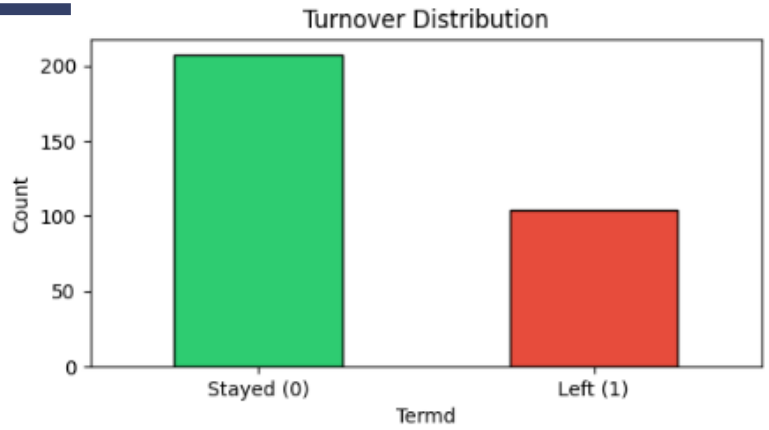
More than 300 employees

Based in Paris, France

HR team of 8

Problem

- **Turnover rate of 33%**
  1 in 3 employees has left
- **Cost of replacing one employee**



How can we balance predictive performance and responsible AI (legal, ethical, and frugal) to anticipate and prevent resignations?

- AI-powered tool that predicts resignation risk per employee
- Explains the top 3 factors driving each score
- Recommends targeted HR actions
- Built on responsible AI : ethical, frugal, explainable, secure

Data & GDPR

Dataset overview:
- 311 rows (employees)
- 36 columns

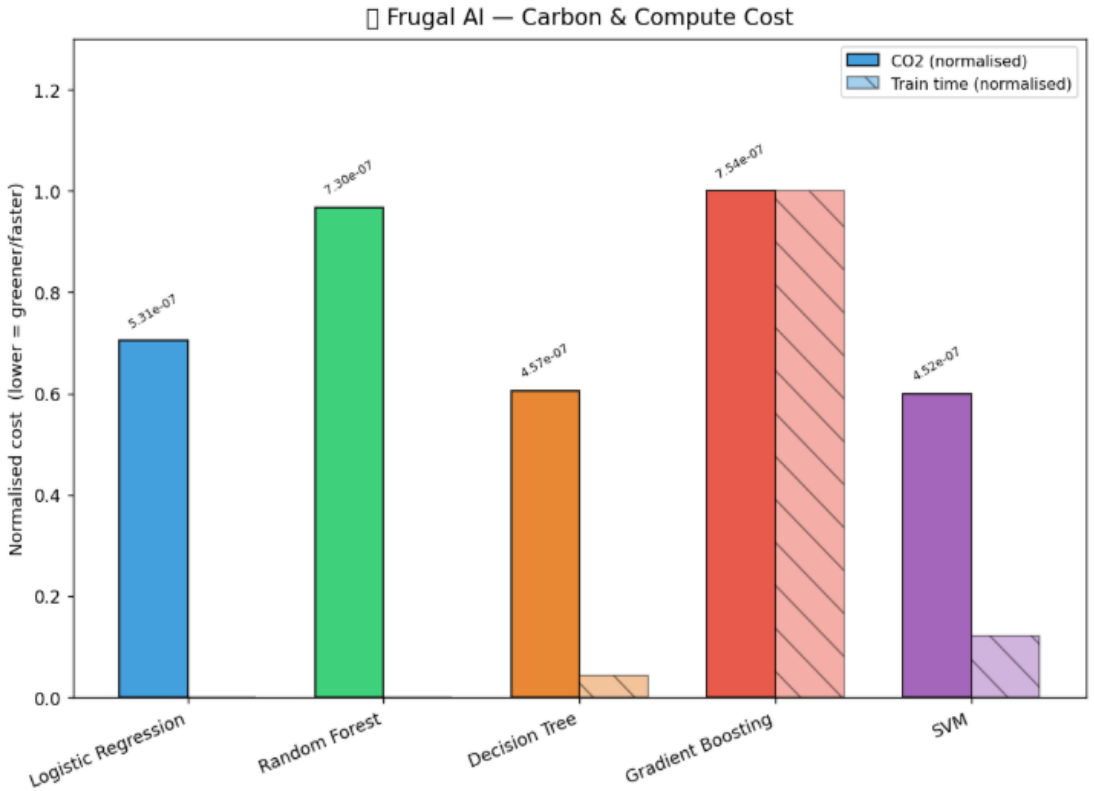
Data privacy
- Anonymization of direct identifiers (Name, DOB, Zip code)
- Ensure strict GDPR compliance

Strict Compliance
- Sensitive data (Sex, Race) fully excluded per French Labor Law (Art. L1132-1)

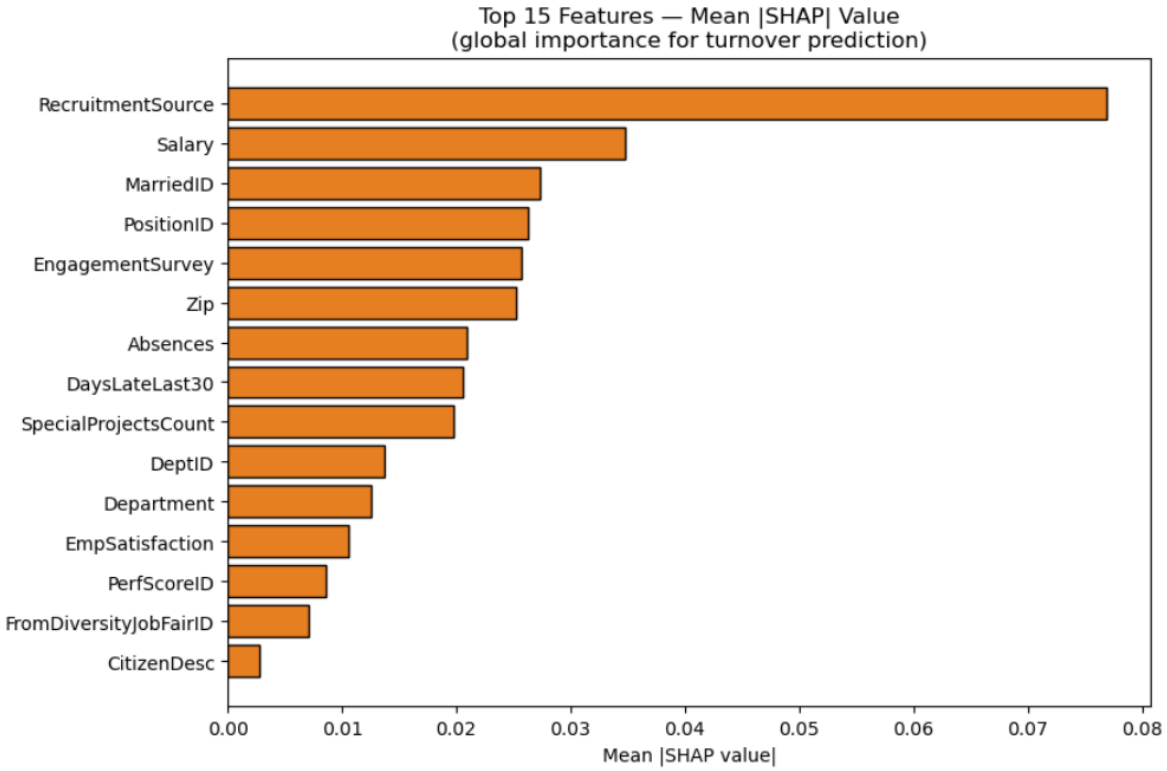
Anonymised dataset:
- 311 rows x 25 columns

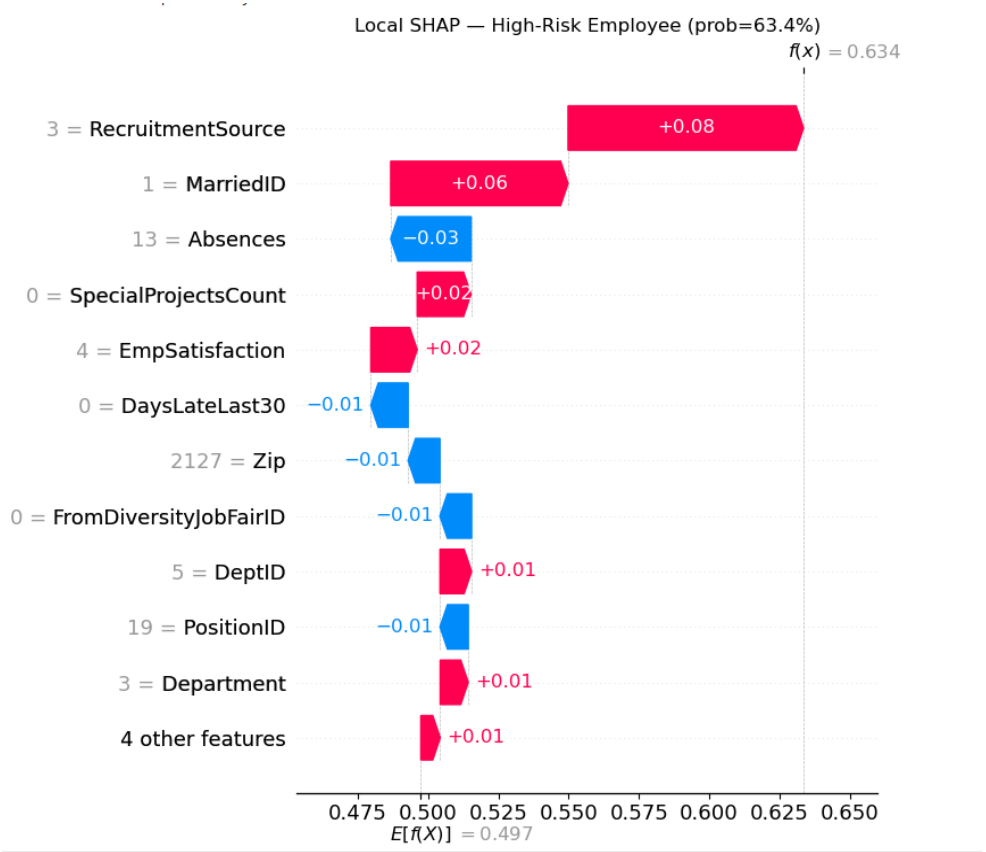
Model Selection (Frugal AI)

Model	AUC	CO2	Selected
Logistic Regression	Good	Lowest	-
Random Forest	Best	Low	✓
Decision Tree	Lower	Lowest	-
Gradient Boosting	Good	Higher	-
SVM	Good	Medium	-



Explainability (SHAP)





Demo

Company-Wide Turnover Risk Overview
311
Total Employees
in dataset
33%
Actual Turnover
historical rate
34
High Risk
prob ≥ 70%
92
Medium Risk
prob 40–70%
185
Low Risk
prob < 40%
Risk Score Distribution
Distribution of Risk Scores
Medium (40%)
High (70%)
Employees
Predicted Turnover Probability
Departure Reason Forecast
Predicted Departure Reasons
(at-risk employees)
Voluntary - Personal
61
Voluntary - Dissatisfaction
45
Voluntary - Work Conditions
6
Voluntary - Better Opportunity
5
Voluntary - Compensation
4
Involuntary - Performance/Conduct
3
Health / Personal
2
Number of employees

Key Takeaways

From Reactive to Proactive HR

Privacy & Ethics by Design

Frugal & Explainable AI

Decision Support